

Confidence Is Not Enough: Evidence-Calibrated, Context-Aware Memory for Long-Horizon AI Agents

Research Thesis — Early-Stage Experimental Framework

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Abstract

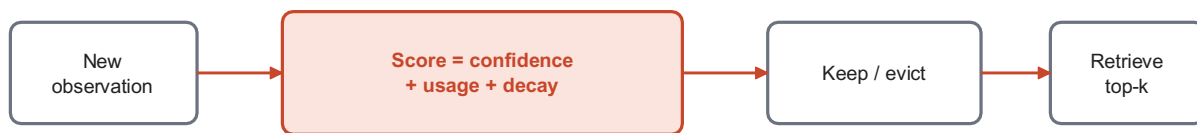
Agents that run for weeks or months need memory that stays correct, not just memory that stays small. Today's agent memories mostly decide what to keep using a handful of scalar parameters: how confident the model was when it wrote the memory, how often the memory has been used, how old it is, and how similar it is to the query. None of these tells the agent whether a memory is actually backed by anything, whether it is still true, whether it applies to the situation at hand, or whether something newer contradicts it. This proposal tests the claim that these parameters are not enough. We propose a memory manager that treats every memory as a revisable belief with attached evidence, provenance, a validity period, a context scope, and links to conflicting memories. Using these, the manager decides whether to store, reinforce, update, weaken, retrieve, or forget each item under a fixed memory budget. The expected outcome is fewer false memories, less stale information, and fewer contradictory retrievals than confidence-based policies at the same memory size and similar retrieval cost. We evaluate this on long-term conversation benchmarks and a controlled stress test where the correct memory state is known at every step.

1. Why Confidence Alone Fails

Most memory pipelines work in three steps: extract facts from the interaction, score each fact (usually with a model-reported confidence), then retain, rank, or evict using that score together with usage counts and age. This is simple and cheap, but each parameter breaks down in a predictable way over long horizons.

Confidence-only memory

keep or evict by confidence, usage, recency

**Evidence-calibrated memory**

decide store, reinforce, update, weaken, or forget per memory

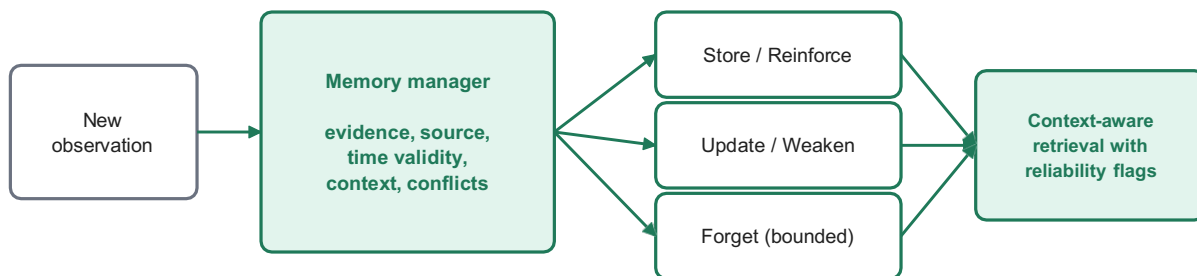


Figure 1. Top: a confidence-based memory scores each item with confidence, usage, and decay, then keeps or evicts it. Bottom: the evidence-calibrated manager looks at evidence, source, validity, context, and conflicts, and picks one of several memory actions.

- **Confidence is poorly calibrated.** A language model’s own confidence about a fact it just extracted is often high even when the extraction is wrong, and it says nothing about whether the user actually said it.
- **Usage creates a feedback loop.** A wrong memory that gets retrieved earns a higher usage score, which makes it more likely to be retrieved again. Popularity ends up standing in for truth.
- **Decay is a blunt proxy for time.** Age-based decay treats a home address, a mood, and a birthday the same way. What matters is how quickly each kind of fact tends to change, and whether it has been replaced.
- **Conflicts go unresolved.** When a user changes a plan or a tool returns new data, old and new versions can sit side by side, and both can be retrieved.
- **Provenance is lost.** A fact the user stated and a fact the agent guessed look identical once stored. Over time the agent can end up treating its own inference as established fact.
- **Scope is ignored.** A preference that holds for one project or persona gets applied where it does not hold.

2. Problem Definition

Let a memory item m carry a belief state that depends on the query time t and the current context c . The system has to make two kinds of decisions:

- **On write:** for each new observation, choose among store as new, reinforce an existing item, update it, weaken it, or forget it.
- **On read:** for each query, choose which items to surface, in what order, and with what reliability flag, given t and c .

We define a reliability score $R(m, t, c)$ that combines calibrated confidence with the net strength of evidence, source trust, validity at time t , match with context c , and the pressure from conflicting items. The objective is to maximise answer correctness while minimising false, stale, and contradictory memory use, subject to a fixed memory budget and a bounded retrieval cost. Sweeping the memory budget produces a reliability–budget curve (Figure 2), which is the main object of study.

The headline measure is the relative reduction in false, stale, and contradictory memory use against a confidence-based policy at the same memory budget.

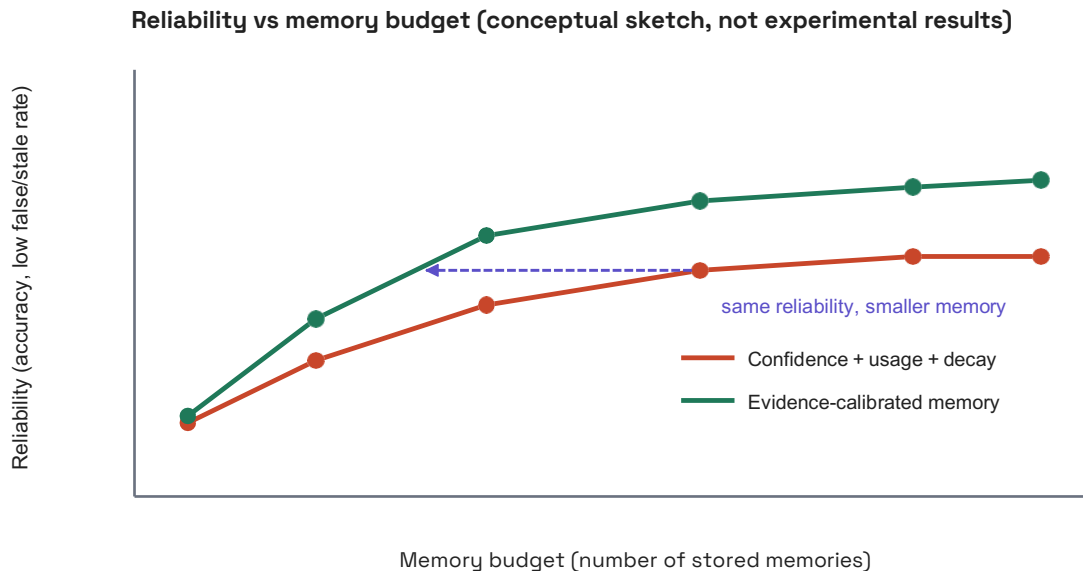
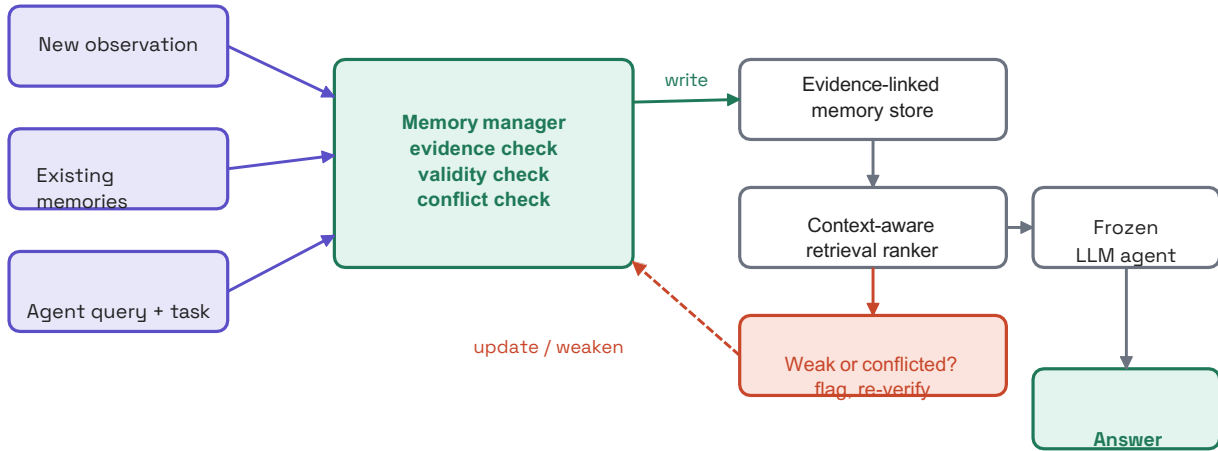


Figure 2. Conceptual sketch of the target result: the evidence-calibrated curve sits above and to the left of the confidence-based curve. This illustrates the hypothesis and is not a measured result.

3. Proposed Architecture

The design adds a memory manager between the agent and its store (Figure 3). The language model stays frozen. The manager itself is a small scoring component plus a few targeted verification calls, so it can be bolted onto an existing agent. It borrows ideas from belief revision and truth maintenance systems: a memory is a belief that must stay justified, and it is retracted or revised when its justification changes.

Evidence-calibrated memory manager on a frozen LLM agent



Training signal: scripted episodes with known fact timelines (what was true, when, and in which context).
No human annotation needed to label memory outcomes.

Figure 3. Memory manager. New observations and existing memories pass through evidence, validity, and conflict checks before any write. At read time, a context-aware ranker filters and flags weak or conflicted memories before the frozen model answers.

3.1 Memory Record

Each memory stores a structured record instead of text plus a score:

Field	What it holds
Content	The fact in normalised form, with an embedding for similarity search.
Calibrated confidence	Extraction confidence mapped to observed correctness on held-out data, so 0.9 actually means about 90%.
Evidence set	Links to the turns, tool outputs, or documents that support or oppose the memory, with a rough strength value.
Provenance	Where it came from: user statement, tool result, agent inference, or third party, plus a trust level for that source type.
Validity period	When the fact was observed and the period it is believed to hold (valid-from, valid-to), plus a volatility class: stable, slow-changing, or fast-changing.
Context scope	The task, project, persona, or topic in which the fact applies.
Conflict links	Edges to memories that disagree with it, with a note on which is newer or better supported.
Usage counters	Retrieval and reinforcement counts, kept apart from evidence so repeated use cannot count as proof.

3.2 Reliability Score

The score is a simple, interpretable combination:

$$R = \text{sigmoid}(w1 * \text{calibrated_confidence} + w2 * \text{net_evidence} + w3 * \text{source_trust} + w4 * \text{validity}(t) + w5 * \text{context_match}(c) - w6 * \text{conflict_pressure}).$$

The weights start as hand-set values, then are learned from the stress-test episodes. Keeping the form simple makes ablations clean: each term can be switched off to measure its contribution.

3.3 Write-Time Manager

For each new observation the manager retrieves related memories and classifies the relation as supports, extends, contradicts, or unrelated. A supporting observation reinforces only if it adds independent evidence; repeating or retrieving the same memory does not. A contradiction that is newer or better supported triggers an update: the old item is weakened, closed off in time, and archived as history rather than deleted. Observations from low-trust or inferred sources are stored as tentative until something corroborates them.

3.4 Context-Aware Retrieval

Candidates are first picked by semantic similarity, then re-ranked using context match, validity at query time, and reliability. Expired, out-of-scope, or defeated memories are dropped or demoted. If a conflict is still unresolved, both versions are passed with a flag, so the agent can ask the user, hedge its answer, or verify, instead of silently choosing.

3.5 Forgetting Under a Fixed Budget

When the store is full, eviction ranks items by expected future usefulness: reliability, how often their context comes up, and whether they can be rebuilt from other memories. Superseded and contradicted items go first, either archived cheaply or merged into a compact summary. Evidence links are stored as short pointers so the extra metadata does not eat the budget.

3.6 Policy Variants

- Rule-based thresholds on evidence strength, source trust, and validity windows (simple baseline).
- Learned scorer trained on synthetic episodes where the true memory state is known at every step.
- Judge-assisted manager that uses a language model for contradiction and entailment checks, behind a cheap pre-filter to limit calls.
- Leave-one-signal-out variants that remove evidence, provenance, validity, context, or conflict handling in turn.

4. Experimental Design

4.1 Datasets and Baselines

Group	Items
Long-term conversation QA	LoCoMo and LongMemEval, covering information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention
Controlled stress test	Synthetic long episodes we generate, with injected false facts, scheduled fact changes, conflicting sources, and context-specific facts, each with a ground-truth timeline
Agentic task streams	A small curated set of long multi-step tasks where earlier tool results or preferences must be recalled much later
Baselines	Full context / no memory; similarity-only memory; recency; recency + usage; a confidence-based survival policy (confidence, usage, and decay) built by us as the main reference; MemGPT-style paging; MemO and A-MEM where reproducible

1.1 Metrics

Category	Metrics
Answer quality	Exact Match, F1, and an LLM-judge or human check on a sample; accuracy on knowledge-update and temporal questions
Memory reliability	False-memory rate, stale-retrieval rate, contradictory-retrieval rate, correct-update rate, false-reinforcement rate, correct-forgetting rate
Calibration	Expected calibration error of the reliability score; quality of abstention when evidence is weak
Efficiency	Memory size, write and read latency, tokens per query, extra manager calls, estimated cost
Trade-off	Reliability–budget curves; reliability at a fixed memory size; memory size at a fixed reliability

1.2 Experiment Plan

ID	Experiment	Purpose
E1	Run baselines on all benchmarks and tag every failure by type	Measure how often confidence-based memory keeps false, stale, or conflicting facts
E2	Evidence and provenance only	Isolate the gain from evidence-aware writing and from tracking sources
E3	Validity period and conflict handling only	Isolate the gain on updates and stale information
E4	Full system: evidence + validity + context + conflict	Main result; budget curves against all baselines
E5	Signal ablation, one signal removed at a time	Find which signals carry the gain
E6	Robustness: other models, noisy or adversarial sources, longer horizons	Check generalisation and find failure cases
E7	Error analysis of wrongly kept, wrongly forgotten, and wrongly updated items	Explain failure modes for the discussion section

1.3 Setup Notes

- Use the same embedding model, retriever, and extraction prompts for every system so differences come from the memory policy alone.
- Run at least two language models: one open-source (7–8B) and one API model, to show the method is not tied to one model.
- Fix a memory budget per run and sweep it over several sizes.
- Log every memory action so any policy can be replayed offline. Use multiple seeds and report confidence intervals.

5. Hypotheses and Expected Contributions

- **H1 (reliability):** at an equal memory budget, evidence-calibrated memory lowers false-memory and stale-retrieval rates against a confidence-based policy (target: at least 30% relative reduction).
- **H2 (updates and conflicts):** explicit validity periods and conflict links raise knowledge-update accuracy and cut contradictory retrieval, with the gap widening as the horizon grows.
- **H3 (signals):** evidence, time, and context signals together beat confidence alone, and beat any single added signal.
- **H4 (efficiency):** memory stays within budget and the extra read and write overhead stays small (target: under 20% added latency).

The central testable hypothesis is: at an equal memory budget, an evidence-calibrated, context-aware memory reduces false, stale, and contradictory retrievals by at least 30% compared with a confidence-based policy, without lowering accuracy on questions that involve no conflict. Expected contributions:

- A memory framework that decides how to store, reinforce, update, retrieve, weaken, and forget using evidence, time, and context, on top of a frozen language model.
- A stress-test generator and a taxonomy of memory failures (false, stale, contradictory, out-of-scope), with ground-truth timelines.
- A signal-by-signal comparison with reliability–budget analysis against confidence-based and published memory systems, with open code.

6. Open Problems and Risks

- **Novelty overlap:** MemGPT, Generative Agents, Mem0, A-MEM, and temporal knowledge-graph memories already cover parts of this space. The angle here is joint control of evidence, validity, context, and conflicts under a fixed budget, compared head-to-head with confidence-based retention. A careful literature check comes first.

- **Evidence can be wrong too:** trusting a source more can reinforce a bad source. Mitigation: source trust is learned and revisable, and E6 tests adversarial inputs.
- **Judge noise and cost:** model-based contradiction checks can be wrong and expensive. Mitigation: a cheap pre-filter, caching, and checking judge accuracy against the scripted ground truth.
- **Synthetic bias:** the stress test could favour the proposed method. Mitigation: report on LoCoMo and LongMemEval too, and include agentic streams not built around the method.
- **Metadata overhead:** richer records cost space and time. Mitigation: compact evidence pointers, with H4 as an explicit constraint.
- **Baseline fairness:** the confidence-based reference is built by us, so we must tune it as carefully as our own method and document every setting.

7. Timeline and Publication Plan

Period	Milestones
Month 1	Literature review; finalise research questions; set up benchmarks, models, and evaluation harness; build the confidence-based reference policy
Month 2	Build the stress-test generator and failure taxonomy; run baselines (E1)
Month 3	Memory record, write-time manager, and retrieval (E2, E3); first results
Month 4	Full system and signal ablations (E4, E5)
Month 5	Robustness and error analysis (E6, E7); refine method
Month 6	Write thesis and paper; release code and stress-test suite

- First target: a workshop paper (agents, memory, or LLM-systems workshops at NeurIPS, ICLR, ACL, or EMNLP) for early feedback.
- Full paper: ACL, EMNLP, NAACL, or an agents-focused venue if the reliability gains are strong.
- Publishability needs: a clear novelty claim, strong baselines (including a tuned confidence-based policy and at least one recent memory system), more than one model, and honest failure analysis.

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